



The inverse care law re-examined: a global perspective

Richard Cookson, Tim Doran, Miqdad Asaria, Indrani Gupta, Fiorella Parra Mujica

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Centre for Health Economics
(Prof R Cookson PhD) and
Department of Health Sciences
(Prof T Doran MD,
F Parra Mujica MSc), University
of York, York, England;
Department of Health Policy,
London School of Economics,
London, England
(M Asaria PhD); Health Policy
Research Unit, Institute of
Economic Growth, Delhi, India
(Prof I Gupta PhD)

Correspondence to:
Prof Richard Cookson, Centre for
Health Economics, University of
York, York YO10 5DD, UK
richard.cookson@york.ac.uk

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An inverse care law persists in almost all low-income and middle-income countries, whereby socially disadvantaged people receive less, and lower-quality, health care despite having greater need. By contrast, a disproportionate care law persists in high-income countries, whereby socially disadvantaged people receive more health care, but of worse quality and insufficient quantity to meet their additional needs. Both laws are caused not only by financial barriers and fragmented health insurance systems but also by social inequalities in care seeking and co-investment as well as the costs and benefits of health care. Investing in more integrated universal health coverage and stronger primary care, delivered in proportion to need, can improve population health and reduce health inequality. However, trade-offs sometimes exist between health policy objectives. Health-care technologies, policies, and resourcing should be subjected to distributional analysis of their equity impacts, to ensure the objective of reducing health inequalities is kept in sight.

Introduction

Tudor Hart coined the term inverse care law (ICL)¹ in 1971 to describe the double injustice that socially disadvantaged people not only tend to be more susceptible to illness than socially advantaged people but also receive less health care.^{2–5} He also noted that social class inequalities in primary care delivery in the UK had been substantially reduced but not eliminated by the introduction of the universal, tax-funded National Health Service (NHS) in 1948.^{3–7} The term has since been widely adopted to describe social inequalities or disparities in health care of all kinds, with similar discussions in other countries,⁸ and similar phrases have been applied to related phenomena, including the “inverse equity hypothesis” (ie, new health interventions are adopted earlier by advantaged populations, thereby initially increasing inequalities)^{9,10} and the “inverse hazard law” (ie, health risks vary inversely with wealth and power).¹¹ The original article has been cited by more than 3500 academic publications, mostly by authors outside the UK, and the annual citation rate continues to rise, with an exponential growth in the general academic literature on health-care inequalities (see appendix 1 pp 3–12).

Our aim in this Health Policy is to re-examine the nature, magnitude, and causes of social inequality of health-care delivery from a global perspective in the light of advances in scientific knowledge since 1971. Heterogeneous reporting makes it hard to compare the magnitude of health-care inequality between studies, and cross-country studies still tend to focus on inequality in health-care utilisation rather than the quality of care. However, in this Health Policy, we make cross-country comparisons of the magnitude of inequality in health-care utilisation among high-income countries and, separately, among low-income and middle-income countries; and we summarise the substantial body of knowledge about the causes of health-care inequality from country-specific studies.

Since 1971, little global progress has been made in tackling the ICL. A complete ICL—defined as a decrease in health-care use with social disadvantage—persists in

almost all low-income and middle-income countries. A disproportionate ICL—defined as an increase in health-care use with social disadvantage but not in proportion to need—persists even in upper-middle and high-income countries with integrated systems of universal health coverage. The complete ICL is largely driven by financial barriers to health care in unregulated health-care markets, and countries with worse governance tend to have larger ICLs. These barriers, and the inequalities associated with them, are reduced under integrated systems of universal health care. However, social inequalities in health-care quality and outcomes persist due to social inequalities in the ability to seek health care (eg, by taking time off work, navigating complex systems, and avoiding discrimination), the ability to benefit from health care (eg, by investing time and resources in following treatment regimens), and the costs and risks of health care (eg, due to multimorbidity and medical workforce shortages).

The ICL reformulated

The ICL was originally formulated as follows: “The availability of good medical care tends to vary inversely with the need for it in the population served. This ICL operates more completely where health care is more exposed to market forces, and less so where such exposure is reduced.”¹ This memorable phrase, punning on inverse square laws from the natural sciences, is an effective communication device that captures imaginations and resonates with people’s experiences. However, the ICL needs to be reformulated in a narrower but more precise way to facilitate empirical investigation of its magnitude and causes from a global perspective.

First, the ICL is solely concerned with differences in health care and need related to social disadvantage (eg, area deprivation, income, social class, ethnicity, gender) and not differences unrelated to social disadvantage. The ICL predicts a negative association between social disadvantage and health-care use among people with the same level of need (horizontal inequity related to social disadvantage), and a positive relationship between social disadvantage and health-care need (vertical

inequity related to social disadvantage). However, among people who share the same level of social disadvantage, the ICL does not necessarily predict that those who need more health care will receive less (ie, vertical inequity unrelated to social disadvantage).

Second, we distinguish inequality in the quantity of health-care resources (eg, workforce, utilisation, expenditure per capita) from inequality in the quality of care (eg, clinical processes and risk-adjusted outcomes). Resource inputs are the structural drivers of quality but they do not provide a complete picture. However, although country-specific indicators of inequality in quality are available, international comparisons are rare¹² because indicators of inequality in quality are typically based on country-specific administrative health datasets that are hard to harmonise and are not widely available in low-income and middle-income countries.¹³

Third, we define the ICL empirically without incorporating the concept of market forces or any other causal mechanism. This definition facilitates a dispassionate scientific approach to investigating different causal mechanisms and how they operate in different social, institutional, and regulatory environments, including the market mechanisms emphasised by Tudor Hart¹⁶ (such as financial barriers for patients and labour market choices by doctors) but also mechanisms that can arise in both market and non-market settings (such as dysfunctional government, non-financial barriers, and unequal costs and benefits of care).

Finally, we distinguish complete and incomplete forms of the ICL. A complete ICL occurs when health-care delivery is inversely related to social disadvantage in absolute terms (measured by resource input per capita). By contrast, an incomplete ICL occurs when health-care delivery is not inversely related to social disadvantage in absolute but in relative terms, after allowing for the additional needs of socially disadvantaged people. Or, paraphrasing Tudor Hart, “the availability of health care is disproportionately related to the need for it in the population served”.¹ We call this incomplete form the disproportionate care law (DCL).

The left panel of figure 1 illustrates the complete ICL, in which socially disadvantaged populations receive less health care than socially advantaged populations. The right-hand panel illustrates the DCL, in which the absolute quantity of health care is no longer inversely related to social disadvantage and might be positively related. However, the availability of health care is still lower in disadvantaged populations when we account for their additional need for health care.

Our reformulated ICL then comes in two versions. The first version is the (complete) ICL. It refers to: (1) social inequality in health—ie, more socially disadvantaged people tend to have worse health than less socially disadvantaged people; and (2) complete social inequality in the quantity of health-care delivery—ie, more socially disadvantaged people tend to receive less health care and

of worse quality. The second version is the DCL. It refers to: (1) social inequality in health—more socially disadvantaged people tend to have worse health than less socially disadvantaged people; and (2) incomplete social inequality in the quantity of health-care delivery—more socially disadvantaged people tend to receive more health care than more socially advantaged people but less as a proportion of need and of worse quality.

The existence of social inequality in health is well established: there are already numerous international reviews of the vast interdisciplinary literature on the social determinants of health^{14–20} and the health determinants of social disadvantage.²¹ In this Health Policy, we therefore focus on social inequality in health-care delivery. For reasons given previously, we focus on quantity of health care in our international comparisons, but we acknowledge that quality is central to health-care delivery and pay close attention to quality in our analysis of causes.

Empirically, it is easier to measure availability than need. Measuring need for health care is a conceptually challenging and data-intensive task, requiring contestable value judgments and detailed data on resource use, morbidity, and other need variables.²² Need variables are almost always incomplete—eg, information might be available on clinical diagnoses, but not on severity, and information might not be available on other determinants of need such as health behaviour, living conditions, family support networks, travel distance to health-care facilities, and local labour market conditions. There are also problems of socially patterned morbidity reporting bias—eg, in low-income and middle-income countries there is often a reverse social gradient in reported morbidity due to serious under-reporting of morbidity in socially disadvantaged populations with little access to diagnostic care and low expectations.²³ There is also scope for ethical disagreement about how much resource is needed, depending on how far scarcity is considered relevant (eg, whether need should be based on effectiveness or

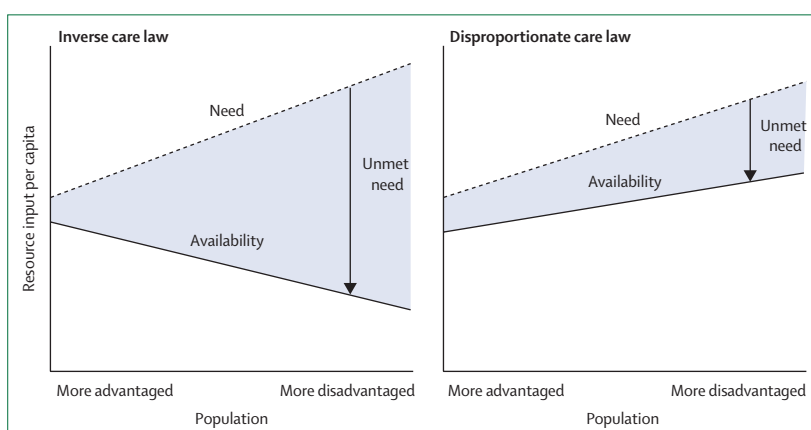


Figure 1: Inverse and disproportionate care laws

Resource input per capita can be measured, for example, as workforce, utilisation, or expenditure, and stratified by age group.

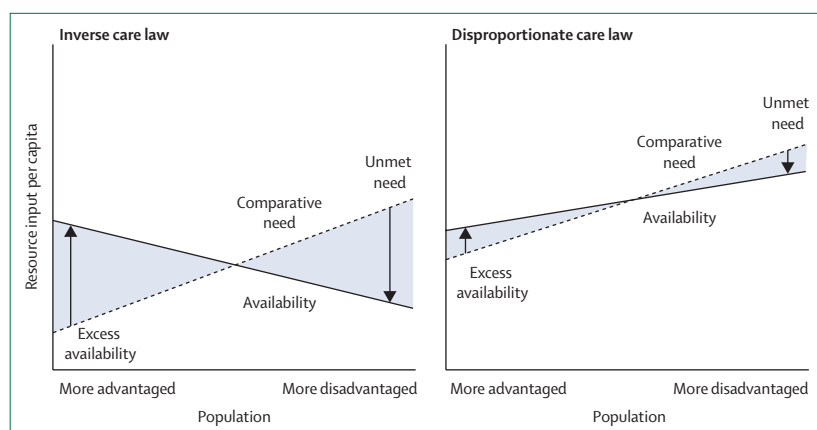


Figure 2: Inverse and disproportionate care laws based on comparative need

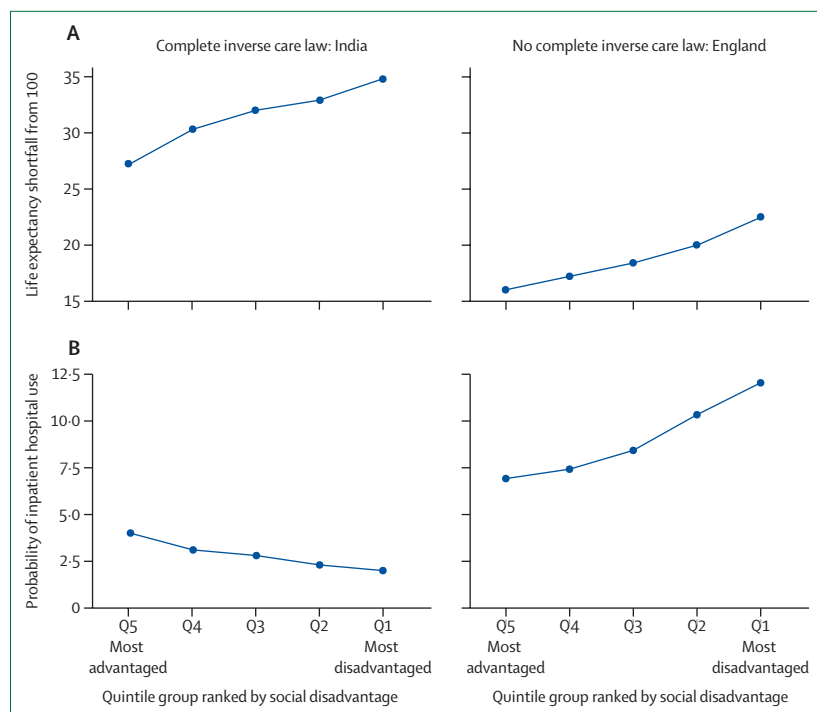


Figure 3: Complete and incomplete inverse care laws in India and England

(A) Life expectancy shortfall by socioeconomic quintile group in India and England. (B) Probability of using inpatient hospital care by socioeconomic quintile group in India and England. We take the inverse of life expectancy by subtracting it from an arbitrary target of 100, and call the result life expectancy shortfall.^{26,27} In both countries the probability of using inpatient hospital care in the past 12 months is measured using self-reported survey data from 2017–18 (using data from the National Sample Survey for India and Understanding Society for England) and includes both elective and emergency admissions.^{28,29} In India, these data exclude childbirth. In England the socioeconomic quintile groups are based on neighbourhood deprivation in (A) and equivalised household annual income in (B), whereas in India they are based on household asset wealth quintile in (A) and equivalised household monthly consumption in (B).

cost-effectiveness) and whether the focus is on expert criteria (normative need), individual perceptions (felt need), or resource use in comparable populations (comparative need).^{24,25}

Perhaps the most fundamental value judgment is whether the focus should be on need for health care that

meets the highest current international standards of quality and effectiveness (normative need) or on need for health care that is cost-effective given current national health-care resource constraints but that might fall short of the highest standards (comparative need).^{24,25} Empirical studies almost always take the latter approach and estimate comparative need on the basis of average resource use in the general population, as illustrated in figure 2.

Figure 3 illustrates the social patterning of ill-health and inpatient hospital utilisation in India and England—polar cases with unusually large (India) and small (England) magnitudes of social inequality in health-care delivery. India exhibits an ICL, with an inverse relationship between health-care utilisation and ill-health.³⁰ In England, however, inpatient hospital utilisation and ill-health are positively related (for both emergency and elective care)³¹ and there is no ICL. Figure 3 does not tell us whether there is a DCL in England for inpatient care, since the amount of need cannot be inferred from life expectancy data alone, but there is evidence of a DCL in England using other indicators—especially for preventive care, primary care, community care, and outpatient care, but also for the utilisation, quality, and risk-adjusted outcomes of specific elective inpatient services such as joint replacement and specific hospital procedures for cardiovascular disease and cancer.^{32–36} This example also raises another conceptual challenge in need measurement, because high levels of inpatient utilisation might partly reflect poor quality in preventive, primary, community, and outpatient care. The actual level of need for inpatient care might thus be higher than the counterfactual ideal level of need that could be achieved if ambulatory care quality were improved. As we discuss in this Health Policy, the magnitudes of ICLs and DCLs vary substantially between different kinds of care and tend to be larger for preventive services than emergency and acute care services.

Cross-country comparisons of social inequality in health-care delivery

Social inequalities in health-care delivery are ubiquitous and studies have documented ICLs and DCLs using diverse measures of health-care quantity (eg, doctors, utilisation, expenditure), quality (eg, patient-reported experiences, clinical processes, risk-adjusted outcomes) and social disadvantage (eg, income, occupational class, education, ethnicity, and gender) at various levels of analysis including geographical (eg, neighbourhood, city, region), organisational (eg, family practice, hospital, insurance plan), and personal (eg, household, family, individual) analyses.^{9,32,37–40} There has also been useful front-line research on patient experience and medical experience in the tradition of Tudor Hart, which is especially useful in assessing and understanding inequality in the quality of care given the limitations of quantitative metrics.^{41–43}

A complete ICL operates within almost all low-income and middle-income countries, which usually have a high private expenditure share or highly fragmented systems of public funding with large urban–rural and employment-related differences in public coverage. Nevertheless, there are exceptions among upper-middle-income countries, including Cuba, which introduced single-payer health coverage in 1959,⁴⁴ and countries that started introducing universal health coverage more recently such as Brazil (since 1988)⁴⁵ and Thailand (since 2002).⁴⁶ In most cases the complete ICL remains when one restricts attention to publicly funded health care alone.^{47–49}

By contrast, a DCL operates in almost all high-income countries, which usually have integrated systems of universal coverage—either single-payer or heavily regulated multi-payer systems with little difference between health insurance plans—and a small (<30%) private expenditure share. A partial exception is the USA, which has a highly fragmented system of means-tested, age-related, and employer-related public subsidies for health care and only approaches universal coverage for people older than 65 years.⁵⁰ Even in the USA, however, there is only a complete ICL for people aged 0–24 years, whereby total health-care expenditure per capita is higher among higher-income citizens than lower-income citizens,⁵¹ although expenditure for disadvantaged groups is heavily skewed towards acute hospital and emergency services and there is a DCL by income and race at all ages and for numerous indicators.^{52–54}

Country-specific studies cannot be used to make wide-ranging global comparisons of the magnitude of social inequality in health-care delivery between countries because of the heterogeneity in variable definitions and inequality metrics, and incomplete reporting of underpinning data—ie, detailed breakdowns by social group as well as unadjusted and adjusted findings—usually precludes the reconstruction of comparable inequality metrics.³²

Nevertheless, some international comparisons of inequality in the quantity of care are possible among low-income and middle-income countries and, separately, among high-income countries. International comparisons among low-income and middle-income countries rely on household survey data from the Demographic and Health Survey and Multiple Indicator Cluster Survey programmes, which are limited to reproductive, maternal, newborn, and child health services and use a household wealth asset index to measure social disadvantage.^{55–57} Using a composite index, these data reveal gaps between the richest and poorest 10% of the population averaging 11.7 percentage points among upper-middle countries, 25.4 percentage points among lower-middle countries, and 27.9 percentage points among low-income countries, with considerable variation ranging from nearly 0 (Thailand) to just over 50 percentage points (Angola; see appendix 1 pp 13–19).⁵⁶ These data focus on

utilisation of basic services needed by almost everyone within a specific target population (eg, whether births are attended by a skilled midwife) without requiring adjustment for morbidity and need. However, they do not provide information about inequalities in later life or among men.

For high-income countries, cross-country household survey data on health-care utilisation and morbidity are available for the full adult population. Information has been compiled by the Organisation for Economic Co-operation and Development (OECD) on the basis of national health survey data for 33 EU and OECD countries between 2014 and 2017.⁵⁸ These comparisons reveal differences in rates of health-care use relative to need of around 0–20%. For example, across countries, the probability of visiting a specialist in the past 12 months was on average 12 percentage points higher in the highest-income quintile group than in the lowest quintile group, after adjusting for age and self-reported morbidity, with a significant gradient in almost all countries. The corresponding inequality gaps for uptake of preventive services such as cervical screening, breast screening, and dental check-ups were even larger—eg, reaching 17 percentage points on average for cervical screening with statistically significant differences in 32 of the 33 countries.^{58–60} However, the average gap in the

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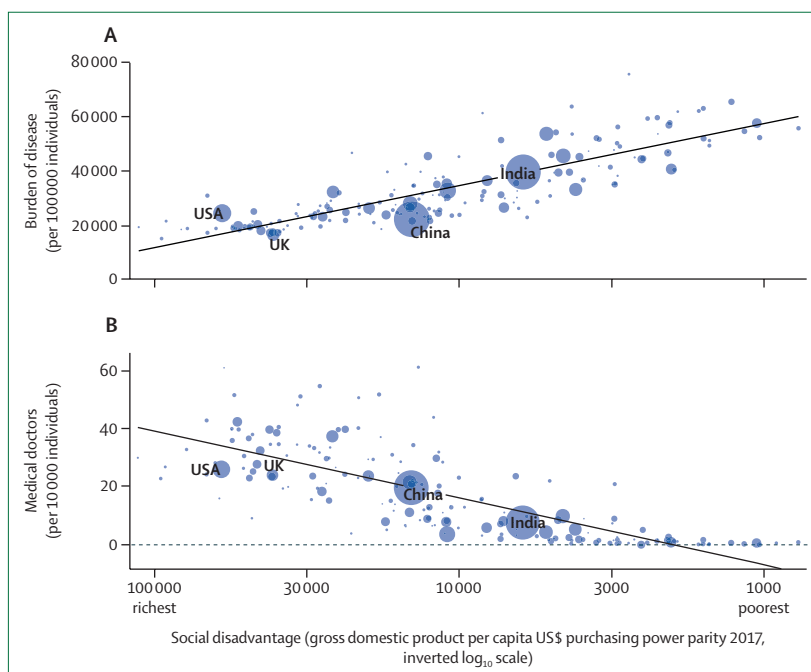


Figure 4: The global inverse care law—social inequality in health and health care between countries
 (A) Burden of disease per capita in 2016 for 180 countries inversely ranked by the log of national income in 2017. (B) Medical doctors per 10 000 population in 2016 for 180 countries inversely ranked by the log of national income in 2017. All data were extracted from the World Bank Database; the original source of burden of disease and health-care access and quality data is the Global Burden of Disease study 2017. The x-axis is the $\log_{10}$ of gross domestic product per capita in constant 2010 US\$. The fitted lines are based on simple linear regressions without population weights, with slopes as follows: (A) $-22\,991$ (95% CI $-25\,285$ to $-20\,696$) and $R^2\,0.69$; and (B) 28.94 (26.71 to 31.18) and $R^2\,0.79$. The raw data can be found in appendix 2.

	Individual factors	Geographical factors	Institutional factors
Social inequality in ability to pay for health care	Wealthier individuals are able to pay more for health care and health insurance	Wealthier communities are able to attract more private investment in health-care facilities	Providers serving wealthier patients are able to charge higher fees and attract larger donations
Social inequality in eligibility for subsidised health care insurance	Ethnic minorities, homeless people, and other vulnerable groups might be less eligible for public subsidy	Advantaged areas are better able than slums, rural areas, and ethnic enclaves to attract public and donor funding	People in higher status occupations might be eligible for better insurance plans with larger public subsidy
Social inequality in non-financial barriers to seeking health care	Advantaged people have greater motivation and ability to seek health care and face less implicit and explicit discrimination	People living in advantaged areas often have shorter journey times to and from high-quality medical facilities	Providers serving advantaged populations are better able to reduce waiting times and other non-financial barriers
Social inequality in barriers to co-investment in health care*	Advantaged patients have better human and social capital and ability to comply with treatment and secure a healthy recovery environment	Lower levels of pollution, crime, and overcrowding decrease the cost of securing a healthy recovery environment	Well funded providers can subsidise co-investment rather than shifting costs to patients
Social inequality in the costs and benefits of health care	Advantaged individuals are better able to avoid multimorbidity and social problems that increase the costs and reduce the benefits of care	Advantaged areas are better able to attract doctors, increasing costs in deprived areas	Providers and specialties serving advantaged people can offer better working conditions and avoid stigma
*The concept of co-investment encompasses various non-health-care resource inputs invested by the patient and their carers to improve the effectiveness of health-care resource inputs. Related concepts include treatment burden, ⁶² long-term care, and self-care, but co-investment also includes broader investments in material and social living conditions that improve treatment compliance and recovery.			
Table: Causes of social inequality in the quantity, quality, and outcomes of health-care delivery			

probability of any visit to the general practitioner in the past 12 months was smaller—at 5 percentage points, only significant for around half of the countries—and the number of repeat visits was higher in lower-income groups than higher-income groups. An international Commonwealth Fund survey found even larger income differentials in self-reported unmet needs for health care in 11 high-income countries.⁶¹ For example, the average self-reported proportion of US citizens reporting that they had skipped needed care because of cost in the past year was 38% with a statistically significant 23% differential between the richest and poorest income groups, whereas in the UK only 10% reported skipping care with only a 5% differential between the richest and poorest groups.⁶¹ However, caution is needed when interpreting country-level rankings within international health-care inequality rankings. Estimates that differ only by a few percentage points are subject not only to statistical uncertainty but also to potential cross-cultural survey response biases in how people in different countries and social groups understand and respond to questions about morbidity and unmet need.

There is social inequality in health care between and within countries. Figure 4 illustrates the between-country ICL; poorer countries tend to have a higher burden of disease and provide fewer medical doctors per 10 000 population than richer countries (see appendix 1 p 19 for the similarly strong association using risk-adjusted mortality as a quality indicator).

The causes of social inequality in health-care delivery

The table lists the main proximal causes of social inequality in health-care delivery within countries.

There are also distal causes of the causes—in particular poor governance and inequalities in wealth, power, human capital, and the conditions of human life—which frustrate efforts to tackle social inequalities in health care. There is a clear cross-country correlation between the magnitude of health-care inequality in low-income and middle-income countries and the quality of governance as measured by the Worldwide Governance Indicators (see appendix 1 p 15).^{63–65} However, there is no simple correlation between income inequality and health-care inequality (see appendix 1 p 16) and relatively unequal countries such as the UK and Thailand can achieve relatively low levels of health-care inequality.

The first direct cause is financial barriers to health care—especially privately funded care. Financial barriers refer to the link between the health care received by an individual and the price paid by that individual, their family, or the employer—what economists refer to as price rationing. The strength of this link is related to the total private share of health expenditure, especially the out-of-pocket share. However, it is not perfectly correlated, especially in high-income countries with heavily regulated insurance systems. For example, in Switzerland nearly 65% of care is privately funded but price rationing is weak because health insurance is mandatory with price and quality controls, means-tested subsidies, and a risk adjustment mechanism to compensate insurers for covering people with pre-existing conditions.^{66,67}

When price rationing is strong, individuals with greater ability to pay will tend to purchase more health care at any non-zero level of need. In India, for example, price rationing of health care via out-of-pocket payments is a major determinant of social inequality in health care.⁶⁸

The effect of price rationing on social inequality in health care depends on income inequality rather than absolute levels of poverty, to some extent because provider (and health service) fees tend to increase in line with income levels of the wealthiest individuals. For example, India halved its absolute poverty rate during a period of rapid economic growth in the 1990s and 2000s,⁶⁹ yet social inequality in health-care utilisation increased (see appendix 1 p 28).

Figure 5 shows the association between social inequality in health-care delivery in low-income and middle-income countries and the share of private health expenditure. The private expenditure share only explains 11% of the variation in health-care inequality—less than the share explained by poor governance—illustrating our main finding that there are many other important causes of social inequality in health care.

A second important cause is fragmented public health insurance due to differential eligibility by ethnicity, caste, housing status, geography, occupational status, and other dimensions of social disadvantage. Socially advantaged communities and the providers that serve them often have greater economic and political power than socially disadvantaged communities, generating large geographical and provider-level inequalities in publicly funded and donor-funded care per capita for health care.⁷² This mechanism is especially powerful in many low-income and middle-income countries, whereby public and donor funding tends to be concentrated on hospital care in urban areas rather than primary care in rural areas.⁷³ It occurs, for example, in Mozambique (figure 5), where most health-care expenditure is funded by external donors,⁷⁴ and Nigeria (figure 5), which has a high share of private expenditure.^{75–77} The UK NHS successfully reduced geographical and institutional inequalities in hospital funding in the 1970s by introducing a geographical resource allocation formula,^{78,79} a policy innovation that has subsequently been adopted elsewhere.⁸⁰

A third important cause is non-financial access barriers. Access to health care often requires the ability to navigate complex health-care systems. Socially advantaged patients have greater health literacy, better digital access, fewer language problems, disabilities, and better social support from family and friends. For example, Indigenous populations often face cultural and linguistic barriers to access in multiethnic societies, lending an additional dimension to stigma and discrimination.⁸¹ Access also requires taking time off work and family duties, which might be easier for socially advantaged individuals,⁸² and effective communication and trust between patient and provider, which can advantage patients with similar cultural affiliations to providers.⁸³ In turn, practitioners' assessments of patient understanding, risk behaviour, and capacity for self-management might be influenced by social status, ethnicity, caste, religion, gender, and condition-related stigma, leading to implicit or explicit

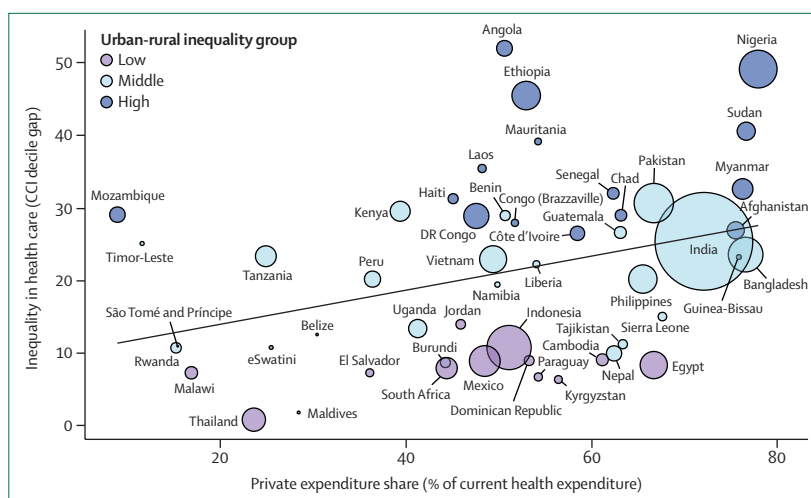


Figure 5: Health-care inequality and private health expenditure in low-income and middle-income countries Percentage point difference in composite index of maternal and child health-care utilisation⁷⁰ is shown between highest and lowest decile group of household asset wealth versus private health-care expenditure share⁷¹ in 62 low-income, lower-middle-income, and upper-middle-income countries, split by low, middle, and high tertile groups of urban-rural inequality in the same index. The source is the WHO Health Equity Assessment Toolkit⁵⁶ and the WHO Global Health Expenditure database. The fitted line is based on a simple linear regression without population weights, with slope 0.24 (95% CI 0.06–0.41) and R^2 0.11. The composite coverage index is a weighted score reflecting receipt of eight reproductive, maternal, newborn, and child health interventions along the continuum of care: demand for family planning satisfied (modern methods); antenatal care received (at least four visits); births attended by skilled health personnel; Bacillus Calmette–Guérin immunisation coverage among 1-year-olds; measles immunisation coverage among 1-year-olds; diphtheria tetanus toxoid and pertussis immunisation coverage among 1-year-olds; children younger than 5 years with diarrhoea receiving oral rehydration therapy and continued feeding; and children younger than 5 years with pneumonia symptoms taken to a health facility. The raw data can be found in appendix 2. CCI=composite coverage index.

discrimination favouring more socially advantaged groups and more informed and effective decision making.^{84–88}

The fourth mechanism is unequal ability to co-invest in health care that improves health-care effectiveness. Socially advantaged people are better able to comply with long-term treatment and lifestyle advice, facilitate communication and care coordination between providers, and provide themselves with a healthy recovery environment and rehabilitation services such as physiotherapy.³² The intergenerational transmission of social inequality via childhood development plays a large role in this mechanism.^{89–93} The human capital, social capital, and cultural capital acquired during childhood has many benefits later in life, and one of them is a greater ability to co-invest in health care. Social inequality in the need for and cost of co-investment can also operate at geographical levels—eg, disadvantaged people living in polluted areas are particularly susceptible to the consequences of pollution, including coping costs and poorer health outcomes.⁸² At the institutional level, well funded providers with better facilities and access to support can transfer less of the effective cost of care onto patients.

The fifth mechanism is social inequality in the costs and benefits of health-care delivery. The concentration of comorbidity, social problems, and psychological distress

in more deprived populations increases the cost of providing care,⁹⁴ which can result in quality being compromised or costs being shifted onto patients.⁹⁵ Additionally, medical students, who are disproportionately drawn from affluent urban communities,^{96,97} tend to return to practice in similar communities to those from which they are drawn, leaving rural and deprived communities struggling to recruit doctors.^{98–100} It can also be hard to attract providers to some rural areas and slums because of violence (eg, gangs and militias) and poor infrastructure. Experienced clinical staff are also lured away from the public to the private sectors by higher salaries and access to better specialist training and facilities.^{100–104} Greater distance from health system infrastructures also reduces the availability of services and increases the costs of providing health care to more distant and deprived communities.¹⁰⁵ Finally, methods of quality assessment and health-care provider reimbursement, including financial and reputational incentive schemes, often disadvantage institutions serving socially disadvantaged populations for whom targets are more challenging to achieve.¹⁰⁶

The causes of social inequality in health care between countries have been less intensively studied. However, it is known that international labour market forces continue to attract well trained doctors to high-income countries and away from low-income and middle-income countries,¹⁰⁷ and that international intellectual property regulations such as the World Trade Organization's TRIPS agreement restrict affordable access to medicines in low-income and middle-income countries.¹⁰⁸

Discussion

A complete ICL continues to operate in almost all low-income and middle-income countries—ie, health-care workforce, utilisation, and expenditure per capita are inversely related to social disadvantage—although the magnitude varies considerably. By contrast, an incomplete ICL operates in all upper-middle-income and high-income countries with integrated systems of universal health coverage, although again the magnitude varies substantially. In these countries, absolute health-care expenditure is positively associated with social disadvantage, especially but not only in relation to emergency hospital care. However, resource input as a proportion of need is still inversely related to social disadvantage, as is the quality of care—a so-called DCL.

The main direct causes of a complete ICL are barriers to financial access (out-of-pocket fees and insurance premiums) that link health-care delivery with the ability to pay as well as public and donor funding mechanisms that link funding to economic and political power rather than need. A complete ICL can exist in countries that claim to be implementing universal health coverage but in reality have fragmented systems of health insurance that vary enormously by plan and region.¹⁰⁹ In such settings, expansions of coverage that leave poor and rural communities

behind can initially increase social inequality in health care (the inverse equity hypothesis).^{10,110}

However, there are several other important mechanisms that cause social inequalities in health-care delivery and generate DCLs even in countries with integrated systems of universal health coverage. These mechanisms include inequalities in the ability to seek health care, inequalities in the ability to co-invest in health care, and inequalities in the costs and risks of health-care delivery. Even if price rationing and geographical inequalities in public funding are virtually eliminated, social inequalities in health care remain.⁵⁸ These additional factors can also give rise to intervention-generated inequalities,^{10,111} whereby well intentioned, cost-effective interventions increase health inequalities because socially advantaged individuals are better able to seek, co-invest, and benefit from them. This potential for unintentionally increasing inequalities can sometimes also generate difficult trade-offs between efficiency (improving total health) and equity (reducing health inequalities).^{112,113} For example, health care might be more cost-effective for socially advantaged individuals who are better able to seek and co-invest in care, who have fewer social problems and risk factors that increase costs and worsen outcomes, and who do not live in disadvantaged areas that struggle to recruit doctors.⁶

For Tudor Hart, introducing integrated universal health coverage was only the first step; universal health coverage needed to be strengthened to remove the remaining social inequalities in health care. Since Tudor Hart's original analysis, some limited progress has been made towards introducing universal health coverage in low-income and middle-income countries, reducing the magnitude of their complete ICLs,¹¹⁴ and in some cases, such as Brazil⁴⁵ and Thailand,⁴⁶ eliminating it. Furthermore, the general global trend has been towards a higher share of public expenditure on health care—even in the USA, where the share of government expenditure is about 58% (or 85% including compulsory private insurance). Overall, however, Tudor Hart would be disappointed at the insufficient global progress over the past 50 years in tackling the ICL in both its complete and incomplete forms. But he might not be surprised: he was well aware that large structural inequalities in wealth and power and poor governance make it hard for societies to develop and strengthen progressive health-care funding systems whereby the rich and the healthy subsidise the poor and the sick. And he also emphasised the political role of excessively large and powerful private sector business interests in lobbying governments, capturing regulators, and distorting national and international policy making. There is hope for more rapid progress in the coming decades, however. Despite the disruption to global health care caused by the COVID-19 pandemic,¹¹⁵ and the ongoing damage to economic growth and public finances, the large-scale collective responses might foster social solidarity and greater public awareness of inequalities in health and health care.

Tudor Hart was particularly concerned to reduce social inequalities in primary care, because achieving such reductions could play a substantial role in decreasing wider inequalities in health from a population perspective, especially but not only in countries without universal primary care.^{116–118} This approach also aligns well with a health economic perspective, according to which scarce resources should be prioritised towards highly cost-effective services—such as primary care, community care, preventive care, and basic surgery—that deliver the largest returns to population health.¹¹⁹ However, strengthening primary care¹²⁰ and improving the effectiveness and efficiency of all health services are necessary but not sufficient to tackle inequalities in health. We also need better information about the health inequality effects of health-care decisions.

Attaining this information will require a reshaping of the national and international research infrastructure for health data collection and research to address equity as well as effectiveness and efficiency.¹²¹ Tudor Hart was an advocate of data gathering and an early adopter of clinical computing systems but might not have anticipated the role information would play in the 21st century in perpetuating, rather than reducing, inequalities. Statistics are the eyes of the state, but the state has a blinkered view when it comes to health inequality impacts. Public decision making still prioritises effectiveness and efficiency over equity, relying on analytical approaches that measure averages rather than social distributions.¹²¹ Health inequality problems are sometimes described,¹²² but little actionable intelligence is provided about solutions: health technologies, programmes, and policies are still routinely evaluated without any serious attempt to quantify their equity effects on social inequalities in health and health care.¹²³ All too often, health technology assessments use inflated value-for-money thresholds^{124,125} and pay no attention to the social distribution of opportunity costs, shifting attention away from reducing health-care inequality.

Resource-constrained decision makers on the path to universal health coverage might face hard choices between covering more people, covering more services, reducing user fees, and funding programmes to tackle non-financial barriers to access, such as discrimination and digital divides.¹²⁶ Different choices will have different effects on effectiveness, efficiency, and equity. Given resource constraints there are also broader social choices about how much to spend on health care versus other social programmes—education, employment, social protection, and so on—that might be more cost-effective ways of improving health and reducing health inequalities.¹²⁷

The ICL thus has radical, transformative research implications both for health service researchers—who need to pay more attention to equity—and for health equity researchers—who need to pay more attention to policy trade-offs. By bringing the two perspectives

together, health-care decisions can be informed by evidence about their effects on equity as well as their effectiveness and efficiency. Methods of distributional cost-effectiveness analysis are now available to make equity a measurable endpoint of health technology and health policy assessment.^{128,129} If decision makers are serious about tackling health inequalities, they should start using these methods routinely to assess the health equity impacts of health technologies and policies.

Contributors

All authors contributed to the study design, literature search, data interpretation, and writing. RC coordinated the study team and led the writing up. FPM collected and analysed the data and created the figures. IG provided additional previously unpublished data on health-care inequality in India from the National Sample Survey Round 75. FPM and RC accessed and verified the underlying data.

Declaration of interests

RC, TD, and FPM declare support from the Wellcome Trust during the conduct of the study. RC is a member of the NHS Advisory Committee on resource allocation. All other authors declare no competing interests.

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References

- Hart JT. The inverse care law. *Lancet* 1971; **1**: 405–12.
- Marmot M. An inverse care law for our time. *BMJ* 2018; **362**: k3216.
- Warren P. Julian Tudor Hart: visionary general practitioner who introduced the concept of the “inverse care law”. *BMJ* 2018; **362**: k3544.
- Cooper K. A review of the literature: does the inverse care law still apply today? 2010. <https://www.sochealth.co.uk/national-health-service/public-health-and-wellbeing/poverty-and-inequality/the-inverse-care-law/a-review-of-the-literature-does-the-inverse-care-law-still-apply-today/> (accessed Nov 22, 2020).
- Le Grand J. The strategy of equality. London: George Allen and Unwin, 1982.
- Watt G. The inverse care law today. *Lancet* 2002; **360**: 252–54.
- Watt G. The inverse care law revisited: a continuing blot on the record of the National Health Service. *Br J Gen Pract* 2018; **68**: 562–63.
- Sengupta A. Universal health coverage: beyond rhetoric. https://www.municipalservicesproject.org/sites/municipalservicesproject.org/files/publications/OccasionalPaper20_Sengupta_Universal_Health_Coverage_Beyond_Rhetoric_Nov2013_0.pdf (accessed Jan 20, 2021).
- Victora CG, Joseph G, Silva ICM, et al. The inverse equity hypothesis: analyses of institutional deliveries in 286 national surveys. *Am J Public Health* 2018; **108**: 464–71.
- Victora CG, Vaughan JP, Barros FC, Silva AC, Tomasi E. Explaining trends in inequities: evidence from Brazilian child health studies. *Lancet* 2000; **356**: 1093–98.
- Krieger N, Chen JT, Waterman PD, et al. The inverse hazard law: blood pressure, sexual harassment, racial discrimination, workplace abuse and occupational exposures in US low-income black, white and Latino workers. *Soc Sci Med* 2008; **67**: 1970–81.
- Arsenault C, Jordan K, Lee D, et al. Equity in antenatal care quality: an analysis of 91 national household surveys. *Lancet Glob Health* 2018; **6**: e1186–95.
- Cookson R, Asaria M, Ali S, Shaw R, Doran T, Goldblatt P. Health equity monitoring for healthcare quality assurance. *Soc Sci Med* 2018; **198**: 148–56.

- 14 Marmot M, Friel S, Bell R, Houweling TA, Taylor S. Closing the gap in a generation: health equity through action on the social determinants of health. *Lancet* 2008; **372**: 1661–69.
- 15 Marmot M, Allen J, Bell R, Bloomer E, Goldblatt P. WHO European review of social determinants of health and the health divide. *Lancet* 2012; **380**: 1011–29.
- 16 Pan American Health Organization, WHO. Just societies: health equity and dignified lives. Report of the commission of the Pan American Health Organization on equity and health inequalities in the Americas. 2019. <https://iris.paho.org/handle/10665.2/51571> (accessed Nov 7, 2020).
- 17 Adler NE, Stewart J. Preface to the biology of disadvantage: socioeconomic status and health. *Ann NY Acad Sci* 2010; **1186**: 1–4.
- 18 Blas E, Kurup S, WHO. Equity, social determinants and public health programmes. 2010. https://apps.who.int/iris/bitstream/handle/10665/44289/9789241563970_eng.pdf?sequence=1&isAllowed=y (accessed Jan 20, 2021).
- 19 Cash-Gibson L, Rojas-Gualdrón DF, Pericás JM, Benach J. Inequalities in global health inequalities research: a 50-year bibliometric analysis (1966–2015). *PLoS One* 2018; **13**: e0191901.
- 20 Collyer TA, Smith KE. An atlas of health inequalities and health disparities research: “How is this all getting done in silos, and why?”. *Soc Sci Med* 2020; **264**: 113330.
- 21 O'Donnell O, Van Doorslaer E, Van Ourti T. Health and inequality. In: Handbook of income distribution. Amsterdam: Elsevier, 2015: 1419–533.
- 22 O'Donnell O, Doorslaer E Van. Analyzing health equity using household survey data. Washington: World Bank Institute, 2008.
- 23 Sen A. Health: perception versus observation. *BMJ* 2002; **324**: 860–61.
- 24 Culyer AJ. Need: the idea won't do but we still need it. In: Cookson R, Claxton K, eds. The humble economist: Tony Culyer on health, health care and social decision making. York: University of York and Office of Health Economics, 2012: 133–52.
- 25 Bradshaw J. Taxonomy of social need. In: McLachlan G, ed. Problems and progress in medical care: essays on current research: 7th series. London: Oxford University Press, 1972: 71–82.
- 26 Love-Koh J, Asaria M, Cookson R, Griffin S. The social distribution of health: estimating quality-adjusted life expectancy in England. *Value Health* 2015; **18**: 655–62.
- 27 Asaria M, Mazumdar S, Chowdhury S, Mazumdar P, Mukhopadhyay A, Gupta I. Socioeconomic inequality in life expectancy in India. *BMJ Glob Health* 2019; **4**: e001445.
- 28 University of Essex. Understanding Society—the UK Household Longitudinal Study. Waves 1–9, 2009–2018. <https://www.understandingsociety.ac.uk/> (accessed Feb 2, 2021).
- 29 Government of India, Ministry of Statistics & Programme Implementation, National Statistical Office. Key indicators of social consumption in India: health. 2019. http://mospi.nic.in/sites/default/files/publication_reports/KI_Health_75th_Final.pdf (accessed Feb 2, 2021).
- 30 Banerjee S, Roy Chowdhury I. Inequities in curative health-care utilization among the adult population (20–59 years) in India: a comparative analysis of NSS 71st (2014) and 75th (2017–18) rounds. *PLoS One* 2020; **15**: e0241994.
- 31 Asaria M, Doran T, Cookson R. The costs of inequality: whole-population modelling study of lifetime inpatient hospital costs in the English National Health Service by level of neighbourhood deprivation. *J Epidemiol Community Health* 2016; **70**: 990–96.
- 32 Cookson R, Propper C, Asaria M, Raine R. Socio-economic inequalities in health care in England. *Fisc Stud* 2016; **37**: 371–403.
- 33 Stoye G, Zaranko B, Shipley M, McKee M, Brunner EJ. Educational inequalities in hospital use among older adults in England, 2004–2015. *Milbank Q* 2020; **98**: 1134–70.
- 34 Fisher R, Dunn P, Gershlick B, Asaria M, Thorlby R. Level or not? 2020. <https://www.health.org.uk/publications/reports/level-or-not> (accessed Jan 20, 2021).
- 35 Delgadillo J, Asaria M, Ali S, Gilbody S. On poverty, politics and psychology: the socioeconomic gradient of mental healthcare utilisation and outcomes. *Br J Psychiatry* 2016; **209**: 429–30.
- 36 Dixon J, King D, Matosevic T, Clark M, Knapp M. Equity in the provision of palliative care in the UK: review of evidence. 2015. <https://www.mariecurie.org.uk/globalassets/media/documents/policy/campaigns/equity-palliative-care-uk-report-full-lse.pdf> (accessed Feb 2, 2021).
- 37 Bago d'Uva T, Jones AM, van Doorslaer E. Measurement of horizontal inequity in health care utilisation using European panel data. *J Health Econ* 2009; **28**: 280–89.
- 38 Wagstaff A, Bredenkamp C, Buisman LR. Progress on global health goals: are the poor being left behind? *World Bank Res Obs* 2014; **29**: 137–62.
- 39 Galea S, Abdalla SM. COVID-19 pandemic, unemployment, and civil unrest: underlying deep racial and socioeconomic divides. *JAMA* 2020; **324**: 227–8.
- 40 WHO, The World Bank. Tracking universal health coverage: 2017 Global Monitoring Report. 2017. <https://apps.who.int/iris/bitstream/handle/10665/259817/9789241513555-eng.pdf;jsessionid=C29E21005A5692511BE2B70BD2D3C941?se> (accessed Feb 2, 2021).
- 41 Watt G. GPs at the deep end. *Br J Gen Pract* 2011; **61**: 66–67.
- 42 Sturgiss E, Tait PW, Douglas K, Chew J-I, Baglow S, Watt G. GPs at the Deep End: identifying and addressing social disadvantage wherever it lies. *Aust J Gen Pract* 2019; **48**: 811–13. <https://search.informit.com.au/documentSummary;dn=729095236713033;res=IELHEA> (accessed Jan 7, 2021).
- 43 Mathers CD. History of global burden of disease assessment at the World Health Organization. *Arch Public Health* 2020; **78**: 77.
- 44 Atun R, de Andrade LOM, Almeida G, et al. Health-system reform and universal health coverage in Latin America. *Lancet* 2015; **385**: 1230–47.
- 45 Barbosa E C, Cookson R. Multiple inequity in health care: an example from Brazil. *Soc Sci Med* 2019; **228**: 1–8.
- 46 Tangcharoensathien V, Patcharanarumol W, Kulthanmanusorn A, Saengruang N, Kosiyaporn H. The political economy of UHC reform in Thailand: lessons for low- and middle-income countries. *Heal Syst Reform* 2019; **5**: 195–208.
- 47 Bowser D, Patenaude B, Bhawalkar M, Duran D, Berman P. Benefit incidence analysis in public health facilities in India: utilization and benefits at the national and state levels. *Int J Equity Health* 2019; **18**: 13.
- 48 McIntyre D, Ataguba JE. How to do (or not to do) ... a benefit incidence analysis. *Health Policy Plan* 2011; **26**: 174–82.
- 49 Asante A, Price J, Hayen A, Jan S, Wiseman V. Equity in health care financing in low- and middle-income countries: a systematic review of evidence from studies using benefit and financing incidence analyses. *PLoS One* 2016; **11**: e0152866.
- 50 Tikkanen R, Osborn R, Mossialos E, Djordjevic A, Wharton GA. United States. 2020. <https://www.commonwealthfund.org/international-health-policy-center/countries/united-states> (accessed Nov 22, 2020).
- 51 French E, Kelly E. Medical spending around the developed world. *Fisc Stud* 2016; **37**: 327–44.
- 52 Fiscella K, Franks P, Gold MR, Clancy CM. Inequality in quality: addressing socioeconomic, racial, and ethnic disparities in health care. *JAMA* 2000; **283**: 2579–84.
- 53 Fiscella K, Sanders MR. Racial and ethnic disparities in the quality of health care. *Annu Rev Public Health* 2016; **37**: 375–94.
- 54 Agency for Healthcare Research and Quality. 2018 National Healthcare Quality and Disparities Report. 2020. <https://www.ahrq.gov/research/findings/nhqrdr/nhqrdr18/index.html> (accessed Jan 20, 2021).
- 55 Gwatkin DR. Trends in health inequalities in developing countries. *Lancet Glob Health* 2017; **5**: e371–72.
- 56 WHO. Health Equity Assessment Toolkit (HEAT): software for exploring and comparing health inequalities in countries. Built-in database edition. Version 4.0. 2020. <https://portal.who.int/heat/> (accessed Nov 6, 2020).
- 57 Victora CG, Barros AJD, França GVA, da Silva ICM, Carvajal-Velez L, Amouzou A. The contribution of poor and rural populations to national trends in reproductive, maternal, newborn, and child health coverage: analyses of cross-sectional surveys from 64 countries. *Lancet Glob Health* 2017; **5**: e402–07.
- 58 OECD. Health for Everyone? 2019. https://www.oecd-ilibrary.org/social-issues-migration-health/health-for-everyone_3c8385d0-en (accessed Jan 20, 2021).
- 59 McKinnon B, Harper S, Moore S. Decomposing income-related inequality in cervical screening in 67 countries. *Int J Public Health* 2011; **56**: 139–52.
- 60 Carrieri V, Wuebker A. Assessing inequalities in preventive care use in Europe. *Health Policy* 2013; **113**: 247–57.

- 61 Doty MM, Tikkanen RS, FitzGerald M, Fields K, Williams RD. Income-related inequality in affordability and access to primary care in eleven high-income countries. *Health Aff (Millwood)* 2021; **40**: 113–20.
- 62 Dobler CC, Harb N, Maguire CA, Armour CL, Coleman C, Murad MH. Treatment burden should be included in clinical practice guidelines. *BMJ* 2018; **363**: k4065.
- 63 Wehrmeister FC, da Silva ICM, Barros AJD, Victora CG. Is governance, gross domestic product, inequality, population size or country surface area associated with coverage and equity of health interventions? Ecological analyses of cross-sectional surveys from 80 countries. *BMJ Glob Health* 2017; **2**: e000437.
- 64 World Bank. Worldwide governance indicators. 2019. <https://info.worldbank.org/governance/wgi/> (accessed Jan 8, 2021).
- 65 Wagstaff A, Bilger M, Buisman L, Breidenkamp C. Who benefits from government health spending and why? A global assessment. 2014. <https://papers.ssrn.com/abstract=2500586> (accessed Nov 24, 2020).
- 66 Reinhardt UE. The Swiss health system: regulated competition without managed care. *JAMA* 2004; **292**: 1227–31.
- 67 Tikkanen R, Osborn R, Mossialos E, Djordjevic A, Wharton GA. Switzerland. 2020. <https://www.commonwealthfund.org/international-health-policy-center/countries/switzerland> (accessed Nov 22, 2020).
- 68 Dwivedi R, Pradhan J. Does equity in healthcare spending exist among Indian states? Explaining regional variations from national sample survey data. *Int J Equity Health* 2017; **16**: 15.
- 69 Himanshu. Widening gaps: India inequality report 2018—Oxfam India. 2018. https://www.oxfamindia.org/sites/default/files/WideningGaps_IndiaInequalityReport2018.pdf (accessed Feb 2, 2021).
- 70 Wehrmeister FC, Barros AJD, Hosseini AR, Boerma T, Victora CG. Measuring universal health coverage in reproductive, maternal, newborn and child health: an update of the composite coverage index. *PLoS One* 2020; **15**: 1–10.
- 71 WHO. Global Health Expenditure Database. 2020. <http://apps.who.int/nha/database/Home/Index/en> (accessed Nov 6, 2020).
- 72 Singh N. Decentralization and public delivery of health care services in India. *Health Aff (Millwood)* 2008; **27**: 991–1001.
- 73 Joseph G, da Silva ICM, Barros AJD, Victora CG. Socioeconomic inequalities in access to skilled birth attendance among urban and rural women in low-income and middle-income countries. *BMJ Glob Health* 2018; **3**: e000898.
- 74 Pfeiffer J, Chapman RR. NGOs, austerity, and universal health coverage in Mozambique. *Global Health* 2019; **15** (suppl 1): 0.
- 75 Nwosu CO, Ataguba JE. Socioeconomic inequalities in maternal health service utilisation: a case of antenatal care in Nigeria using a decomposition approach. *BMC Public Health* 2019; **19**: 1493.
- 76 Adeyanju O, Tubefu S, Ensor T. Socio-economic inequalities in access to maternal and child healthcare in Nigeria: changes over time and decomposition analysis. *Health Policy Plan* 2017; **32**: 1111–18.
- 77 Faye CM, Wehrmeister FC, Melesse DY, et al. Large and persistent subnational inequalities in reproductive, maternal, newborn and child health intervention coverage in sub-Saharan Africa. *BMJ Glob Health* 2020; **5**: e002232–002232.
- 78 Smith PC. Formula funding of public services: an economic analysis. *Oxf Rev Econ Policy* 2003; **19**: 301–22.
- 79 Smith PC. Resource allocation and purchasing in the health sector: the English experience. *Bull World Health Organ* 2008; **86**: 884–88.
- 80 McGuire F, Revill P, Twea P, Mohan S, Manthulu G, Smith PC. Allocating resources to support universal health coverage: development of a geographical funding formula in Malawi. *BMJ Glob Health* 2020; **5**: e002763.
- 81 Tobasia-Hege C, Pinart M, Madeira S, et al. Disrespect and abuse during childbirth and abortion in Latin America: systematic review and meta-analysis. *Rev Panam Salud Publica* 2019; **43**: e36 (in Spanish).
- 82 Gupta S, Das S, Murty MN. Quantifying air pollution vulnerability and its distributional consequences. *Ecol Econ Soc INSEE J* 2019; **2**: 93–125.
- 83 Berkman ND, Sheridan SL, Donahue KE, et al. Health literacy interventions and outcomes: an updated systematic review. *Evid Rep Technol Assess (Full Rep)* 2011; 1–941.
- 84 Balsa AI, McGuire TG, Meredith LS. Testing for statistical discrimination in health care. *Health Serv Res* 2005; **40**: 227–52.
- 85 Lewis TT, Cogburn CD, Williams DR. Self-reported experiences of discrimination and health: scientific advances, ongoing controversies, and emerging issues. *Annu Rev Clin Psychol* 2015; **11**: 407–40.
- 86 Rivenbark JG, Ichou M. Discrimination in healthcare as a barrier to care: experiences of socially disadvantaged populations in France from a nationally representative survey. *BMC Public Health* 2020; **20**: 31.
- 87 Durand M-A, Carpenter L, Dolan H, et al. Do interventions designed to support shared decision-making reduce health inequalities? A systematic review and meta-analysis. *PLoS One* 2014; **9**: e94670.
- 88 Ben J, Cormack D, Harris R, Paradies Y. Racism and health service utilisation: a systematic review and meta-analysis. *PLoS One* 2017; **12**: e0189900.
- 89 Heckman JJ. The economics, technology, and neuroscience of human capability formation. *Proc Natl Acad Sci USA* 2007; **104**: 13250–55.
- 90 Almond D, Currie J, Duque V. Childhood circumstances and adult outcomes: Act II. *J Econ Lit* 2018; **56**: 1360–446.
- 91 Nelson CA, Scott RD, Bhutta ZA, Harris NB, Danese A, Samara M. Adversity in childhood is linked to mental and physical health throughout life. *BMJ* 2020; **371**: m3048.
- 92 Braveman P, Barclay C. Health disparities beginning in childhood: a life-course perspective. *Pediatrics* 2009; **124** (suppl 3): S163–75.
- 93 Conti G, Mason G, Poupakis S. Developmental origins of health inequality. 2019. https://econpapers.repec.org/paper/ifsifsewp/19_2fl7htm (accessed Jan 20, 2021).
- 94 McPhail SM. Multimorbidity in chronic disease: impact on health care resources and costs. *Risk Manag Healthc Policy* 2016; **9**: 143–56.
- 95 Sum G, Hone T, Atun R, et al. Multimorbidity and out-of-pocket expenditure on medicines: a systematic review. *BMJ Glob Health* 2018; **3**: e000505.
- 96 Steven K, Dowell J, Jackson C, Guthrie B. Fair access to medicine? Retrospective analysis of UK medical schools application data 2009–2012 using three measures of socioeconomic status. *BMC Med Educ* 2016; **16**: 11.
- 97 Dowell J, Norbury M, Steven K, Guthrie B. Widening access to medicine may improve general practitioner recruitment in deprived and rural communities: survey of GP origins and current place of work career choice, professional education and development. *BMC Med Educ* 2015; **15**: 1–7.
- 98 Wilson NW, Couper ID, De Vries E, Reid S, Fish T, Marais BJ. A critical review of interventions to redress the inequitable distribution of healthcare professionals to rural and remote areas. *Rural Remote Health* 2009; **9**: 1060.
- 99 Dolea C, Stormont L, Braichet JM. Evaluated strategies to increase attraction and retention of health workers in remote and rural areas. *Bull World Health Organ* 2010; **88**: 379–85.
- 100 Verma P, Ford JA, Stuart A, Howe A, Everington S, Steel N. A systematic review of strategies to recruit and retain primary care doctors. *BMC Health Serv Res* 2016; **16**: 126.
- 101 Sherr K, Mussa A, Chilundo B, et al. Brain drain and health workforce distortions in Mozambique. *PLoS One* 2012; **7**: e35840.
- 102 Basu S, Andrews J, Kishore S, Panjabi R, Stuckler D. Comparative performance of private and public healthcare systems in low- and middle-income countries: a systematic review. *PLoS Med* 2012; **9**: e1001244.
- 103 Mullan F. Doctors for the world: Indian physician emigration. *Health Aff (Millwood)* 2006; **25**: 380–93.
- 104 Huot S, Ho H, Ko A, et al. Identifying barriers to healthcare delivery and access in the Circumpolar North: important insights for health professionals. *Int J Circumpolar Health* 2019; **78**: 1571385.
- 105 Bertolini P, Montanari M, Peragine V. Poverty and social exclusion in rural areas. 2008. https://www.researchgate.net/publication/237103993_POVERTY_AND_SOCIAL_EXCLUSION_IN_RURAL_AREAS (accessed Feb 2, 2021).
- 106 Doran T, Maurer KA, Ryan AM. Impact of provider incentives on quality and value of health care. *Annu Rev Public Health* 2017; **38**: 449–65.
- 107 Mullan F. The metrics of the physician brain drain. *N Engl J Med* 2005; **353**: 1810–18.
- 108 Smith RD, Correa C, Oh C. Trade, TRIPS, and pharmaceuticals. *Lancet* 2009; **373**: 684–91.

- 109 Shroff ZC, Marten R, Ghaffar A, et al. On the path to Universal Health Coverage: aligning ongoing health systems reforms in India. *BMJ Glob Health* 2020; 5: 3801.
- 110 Gwatkin DR, Ergo A. Universal health coverage: friend or foe of health equity? *Lancet* 2011; 377: 2160–61.
- 111 Lorenc T, Petticrew M, Welch V, Tugwell P. What types of interventions generate inequalities? Evidence from systematic reviews. *J Epidemiol Community Heal* 2012; 67: 190–93.
- 112 Williams AH, Cookson RA. Equity-efficiency trade-offs in health technology assessment. *Int J Technol Assess Health Care* 2006; 22: 1–9.
- 113 Fryatt RB. Reaching the poor with health, nutrition and population services: what works, what doesn't, and why. *Int J Epidemiol* 2006; 35: 1109–10.
- 114 Boerma T, Eozenou P, Evans D, Evans T, Kieny M-P, Wagstaff A. Monitoring progress towards universal health coverage at country and global levels. *PLoS Med* 2014; 11: e1001731.
- 115 UN. The Sustainable Development Goals Report. 2020. <https://unstats.un.org/sdgs/report/2020/> (accessed Nov 30, 2020).
- 116 Sheldon T. Vigorous implementation of effective care can reduce inequalities in health. *J Health Serv Res Policy* 2011; 16: 118–20.
- 117 Barr B, Bambra C, Whitehead M. The impact of NHS resource allocation policy on health inequalities in England 2001–11: longitudinal ecological study. *BMJ* 2014; 348: g3231.
- 118 Starfield B, Shi L, Macinko J. Contribution of primary care to health systems and health. *Milbank Q* 2005; 83: 457–502.
- 119 Haines A, Sanders D, Lehmann U, et al. Achieving child survival goals: potential contribution of community health workers. *Lancet* 2007; 369: 2121–31.
- 120 Walley J, Lawn JE, Tinker A, et al. Primary health care: making Alma-Ata a reality. *Lancet* 2008; 372: 1001–07.
- 121 Cookson R, Robson M, Skarda I, Doran T. Equity-informative methods of health services research. *J Health Organ Manag* (in press).
- 122 Doran T, Cookson R. Re-engineering health policy research to measure equity impacts. In: Evans J, Ruane S, Southall H, eds. *Data in society: challenging statistics in an age of globalisation*. Bristol: Policy Press, 2019.
- 123 Horton R. Offline: the error of our health technology assessment ways. *Lancet* 2013; 382: 1318.
- 124 Leech AA, Kim DD, Cohen JT, Neumann PJ. Use and misuse of cost-effectiveness analysis thresholds in low- and middle-income countries: trends in cost-per-DALY studies. *Value Health* 2018; 21: 759–61.
- 125 Woods B, Revill P, Sculpher M, Claxton K. Country-level cost-effectiveness thresholds: initial estimates and the need for further research. *Value Health* 2016; 19: 929–35.
- 126 WHO. Making fair choices on the path to universal health coverage: final report of the WHO consultative group on equity and universal health coverage. 2014 <https://apps.who.int/iris/handle/10665/112671> (accessed Feb 2, 2021).
- 127 Athana S, Gibson A, Halliday J. The medicalisation of health inequalities and the English NHS: the role of resource allocation. *Health Econ Policy Law* 2013; 8: 167–83.
- 128 Cookson R, Griffin S, Norheim OF, Culyer AJ. *Distributional cost-effectiveness analysis: quantifying health equity impacts and trade-offs*. Oxford: Oxford University Press, 2020.
- 129 Cookson R, Griffin S, Norheim OF, Culyer AJ, Chalkidou K. Distributional cost-effectiveness analysis comes of age. *Value Health* 2020; 24: 118–20.

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